



PARAGON
Laboratories

October 28, 2011

Amy Vail
City of Cadillac WWTP
200 N. Lake St.
Cadillac, MI 49601

Subject: Executive Summary for Project Number 180865

Dear Ms. Vail:

Please find the attached biomonitoring report for testing conducted October 06, 2011 through October 13, 2011. The results for this evaluation are:

Test Species	TUa	TUc	MATC
Ceriodaphnia dubia	0	0	>100

The exposure tests were valid since they met the criteria for survival, reproduction, and growth in the controls. Should you have any questions pertaining to this report please let me know.

Sincerely,
Jeannine Canze
Paragon Laboratories, Inc.

Prepared by Paragon Laboratories, Inc. Livonia MI

DEQ

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY - WATER BUREAU CERIODAPHNIA DUBIA CHRONIC TOXICITY TEST REPORT By authority of PA 451 of 1994, as amended.

1. NAME OF FACILITY (on NPDES permit) Cadillac WWTP		2. NPDES PERMIT # MI0020257	
3. RECEIVING WATER (as designated in permit) Clam River		4. OUTFALL 001	5. RECEIVING WATER CONCENTRATION (if known) Clam River
6. TEST LAB (Name and Address) Paragon Laboratories, Inc. 12649 Richfield Ct. Livonia MI 48150			
7. TEST START DATE 10/06/11	8. TEST END DATE 10/13/11	9. AGE RANGE OF ORGANISMS AT TEST START Less Than 24 Hours	10. Report Date 10/28/2011 11:08:32 AM
11. NAME OF PERSON CONDUCTING TEST Jeannine Canze		12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT Paragon Laboratories, Inc. (734) 462-3900	
13. SAMPLE COLLECTION DATES 10/5/2011 7:00:00 AM	14. DATE RECEIVED 10/6/2011 9:50:00 AM	15. DATE OF FIRST USE 10/06/2011	16. ARRIVAL TEMP (°C) Sample 1: 3.0 on ice
10/7/2011 8:00:00 AM	10/8/2011 11:00:00 AM	10/10/2011	Sample 2: 3.3 on ice
10/10/2011 7:15:00 AM	10/11/2011 9:57:00 AM	10/12/2011	Sample 3: 3.0 on ice
17. TOTAL RESIDUAL CHLORINE Sample 1: <0.02	18. AMMONIA (mg/L as N) Sample 1: <0.5	19. pH Sample 1: 7.60	20. Alkalinity Sample 1: 108.4
Sample 2: <0.02	Sample 2: <0.5	Sample 2: 8.10	Sample 2: 170.3
Sample 3: <0.02	Sample 3: <0.5	Sample 3: 7.90	Sample 3: 108.3
21. Hardness Sample 1: 224.3		Sample 2: 271.4	
Sample 3: 233.9			
22. WAS SAMPLE DECHLORINATED? Sample 1: unknown Sample 2: No Sample 3: No		23. DESCRIBE DECHLORINATION (if any) N/A	
24. EFFLUENT SAMPLES WERE COLLECTED: unknown			
25. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded hold time.) N/A			
26. EFFLUENT FILTERED? No		27. STATE MESH SIZE OF FILTER (if filtered) N/A	
28. EFFLUENT SAMPLE TYPE: Sample 1: 24-Hour Composite Sample 2: 24-Hour Composite Sample 3: 24-Hour Composite			29. IDENTIFY THE DILUENT (O1) MHW IDENTIFY THE SECONDARY (O2) N/A
30. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCTION			
CONCENTRATION OF EFFLUENT (%)	O1	5	15
48-HOUR SURVIVAL (%)	100	100	100
7-DAY MEAN REPRODUCTION/FEMALE	21.7	30	27.4
7-DAY MEAN SURVIVAL (%)	100.0	100.0	90.0
31. 48-HOUR LC50 (%)	32. TUa (acute toxic units) 0		
>100			
33. 7-DAY CHRONIC VALUE (%)	34. NOEC	35. LOEC	36. TUc (chronic toxic units)
>100	>100	>100	0

Prepared by Paragon Laboratories, Inc. Livonia MI

TOXICITY EVALUATION
Ceriodaphnia Survival and Reproduction Test

Client: City of Cadillac WWTP

Project Number: 180865

Primary Analyst: Jeannine Canze

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		1	2	3	4	5	6	7	8	9	10			
0%	0*	1	1	1	1	1	1	1	1	1	1	0	1	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	4	2	3	4	3	3	5	2	4	3	2	31	10	3.1
	5	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	6	8	7	10	8	7	7	8	10	12	7	84	10	8.4
	7	12	11	10	9	10	10	8	11	12	9	102	10	10.2
	8											217	1	21.7
	Total	22	21	24	20	20	22	18	25	27	18	217	1	21.7

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		11	12	13	14	15	16	17	18	19	20			
5%	0*	1	1	1	1	1	1	1	1	1	1	0	1	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	4	4	4	5	4	3	7	6	4	3	3	43	10	4.3
	5	13	8	0	10	8	0	0	9	0	0	48	10	4.8
	6	0	0	10	0	0	11	12	0	10	10	53	10	5.3
	7	15	15	17	14	14	15	16	17	16	17	156	10	15.6
	8											300	1	30.0
	Total	32	27	32	28	25	33	34	30	29	30	300	1	30.0

*Neonates less than 24 hours old added to each replicate vessel at start of test. Values are not included in replicate total.

TOXICITY EVALUATION
Ceriodaphnia Survival and Reproduction Test

Client: City of Cadillac WWTP

Project Number: 180865

Primary Analyst: Jeannine Canze

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		21	22	23	24	25	26	27	28	29	30			
15%	0*	1	1	1	1	1	1	1	1	1	1	0	1	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	0	0	0	0	0	0	0	0	0	0	0	9	0.0
	4	0	5	6	6	4	5	5	4	5	6	46	9	5.1
	5	0	7	9	9	8	0	0	10	8	0	51	9	5.7
	6	0	0	0	0	12	13	11	0	12	14	62	9	6.9
	7	0	19	15	12	0	14	13	11	14	17	115	9	12.8
	8											274	1	30.4
	Total	0	31	30	27	24	32	29	25	39	37	274	1	30.4

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		31	32	33	34	35	36	37	38	39	40			
30%	0*	1	1	1	1	1	1	1	1	1	1	0	1	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	4	2	7	5	4	6	5	5	3	5	6	48	10	4.8
	5	0	9	8	9	10	0	10	11	10	0	67	10	6.7
	6	8	0	0	0	0	12	0	0	0	9	29	10	2.9
	7	9	0	12	10	15	15	14	16	16	18	125	9	13.9
	8											269	1	28.3
	Total	19	16	25	23	31	32	29	30	31	33	269	1	28.3

*Neonates less than 24 hours old added to each replicate vessel at start of test. Values are not included in replicate total.

TOXICITY EVALUATION
Ceriodaphnia Survival and Reproduction Test

Client: City of Cadillac WWTP

Project Number: 180865

Primary Analyst: Jeannine Canze

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		41	42	43	44	45	46	47	48	49	50			
60%	0*	1	1	1	1	1	1	1	1	1	1	0	1	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	4	4	6	6	4	7	4	4	5	5	5	50	10	5.0
	5	9	8	0	8	9	0	7	12	8	8	69	10	6.9
	6	0	0	12	0	14	10	0	0	0	0	36	10	3.6
	7	10	0	14	12	0	12	15	12	11	10	96	9	10.7
	8											251	1	26.2
	Total	23	14	32	24	30	26	26	29	24	23	251	1	26.2

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		51	52	53	54	55	56	57	58	59	60			
100%	0*	1	1	1	1	1	1	1	1	1	1	0	1	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	4	5	5	4	5	6	4	5	6	4	4	48	10	4.8
	5	7	9	8	8	10	0	9	8	10	9	78	9	8.7
	6	0	0	0	0	0	8	0	0	0	0	8	8	1.0
	7	0	0	10	11	10	14	12	11	13	14	95	8	11.9
	8											229	1	26.3
	Total	12	14	22	24	26	26	26	25	27	27	229	1	26.3

*Neonates less than 24 hours old added to each replicate vessel at start of test. Values are not included in replicate total.

TOXICITY EVALUATION
Ceriodaphnia Dubia
Routine Chemical and Physical Determination

Client: City of Cadillac WWTP

Project Number: 180865

Primary Analyst: Jeannine Canze

Parameter	Units	Day							
		0	1	2	3	4	5	6	7

0% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	7.8	8.1	8.7	8.2	8.3	8.6	8.2	
pH	S.U.	7.8	7.8	8.2	8.0	8.2	8.0	7.9	
Conductivity	umhos/cm	300	300	280	300	290	300	280	

0% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	8.3	8.0	7.9	8.4	8.1	8.3	8.4	
pH Cerio	S.U.	7.8	8.2	8.3	8.3	8.1	8.2	8.5	
Conductivity Cerio	umhos/cm	320	350	330	300	320	300	320	

5% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	8.0	8.0	8.9	8.0	8.3	8.5	8.3	
pH	S.U.	7.9	8.0	8.2	8.2	8.2	8.1	8.1	
Conductivity	umhos/cm	350	390	350	360	330	430	330	

5% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	8.2	8.2	8.2	8.5	8.1	8.4	8.6	
pH Cerio	S.U.	8.1	8.2	8.4	8.4	8.1	8.3	8.8	
Conductivity Cerio	umhos/cm	370	440	390	370	370	420	370	

15% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	8.2	8.1	9.0	8.1	8.2	8.5	8.3	
pH	S.U.	8.0	8.0	8.2	8.2	8.2	8.1	8.1	
Conductivity	umhos/cm	480	380	450	450	450	440	390	

15% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	8.2	8.1	8.1	8.6	8.1	8.4	8.5	
pH Cerio	S.U.	8.2	8.2	8.4	8.4	8.2	8.3	8.7	
Conductivity Cerio	umhos/cm	490	440	500	460	500	440	440	

TOXICITY EVALUATION
Ceriodaphnia Dubia
Routine Chemical and Physical Determination

Client: City of Cadillac WWTP

Project Number: 180865

Primary Analyst: Jeannine Canzo

Parameter	Units	Day							
		0	1	2	3	4	5	6	7

30% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	8.3	8.1	9.1	8.1	8.1	8.8	8.4	
pH	S.U.	8.0	8.0	8.2	8.2	8.1	8.1	8.1	
Conductivity	umhos/cm	550	500	540	550	570	530	480	

30% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	8.3	8.2	8.1	8.6	8.2	8.4	8.5	
pH Cerio	S.U.	8.3	8.2	8.4	8.4	8.2	8.4	8.7	
Conductivity Cerio	umhos/cm	560	570	590	550	640	520	530	

60% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	8.4	8.0	9.3	8.1	8.0	8.9	8.3	
pH	S.U.	7.9	8.0	8.1	8.1	8.1	8.1	8.1	
Conductivity	umhos/cm	830	770	820	860	850	720	690	

60% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	8.3	8.1	8.0	8.6	8.2	8.4	8.4	
pH Cerio	S.U.	8.3	8.2	8.5	8.5	8.3	8.4	8.7	
Conductivity Cerio	umhos/cm	820	860	860	860	940	680	770	

100% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	9.0	8.0	9.1	8.0	7.8	9.6	8.4	
pH	S.U.	7.9	7.9	8.1	8.1	8.0	8.1	8.0	
Conductivity	umhos/cm	1200	1100	1300	1200	1200	1100	940	

100% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	8.2	8.0	8.0	8.6	8.3	8.4	8.2	
pH Cerio	S.U.	8.4	8.2	8.6	8.5	8.5	8.5	8.6	
Conductivity Cerio	umhos/cm	1100	1200	1300	1200	1400	990	1100	

Prepared by Paragon Laboratories, Inc. Livonia MI

Qualifiers and Comments for Ceriodaphnia Dubia Data

Day 3 Cup 21, natural death.

Day 6 Cup 51, natural death.

Day 6 Cup 52, natural death.

Day 7 Cup 32, natural death.

Day 7 Cup 42, natural death.

RC180865

Title: Ceriodaphnia Reproduction City of Cadillac WWTP 180865
 File: CR180865.TXT Transform: NO TRANSFORMATION

Kolmogorov Test for Normality

D = 0.1421 (p-value = 0.0041)
 D* = 1.1151

Critical D* = 1.035 (alpha = 0.01 , N = 60)
 = 0.895 (alpha = 0.05 , N = 60)

Data FAIL normality test (alpha = 0.01). Try another transformation.

Warning - The first three homogeneity tests are sensitive to non-normality
 and should not be performed with this data as is.

Bartlett's Test for Homogeneity of Variance

Calculated B1 statistic = 21.6475 (p-value = 0.0006)

Data FAIL B1 homogeneity test at 0.01 level. Try another transformation.

Critical B = 15.0863 (alpha = 0.01, df = 5)
 = 11.0705 (alpha = 0.05, df = 5)

Summary Statistics on Data

TABLE 1 of 2

GRP	IDENTIFICATION	N	MIN	MAX	MEAN
1	Control	10	18.0000	27.0000	21.7000
2	5%	10	25.0000	34.0000	30.0000
3	15%	10	0.0000	39.0000	27.4000
4	30%	10	16.0000	33.0000	26.9000
5	60%	10	14.0000	32.0000	25.1000
6	100%	10	12.0000	27.0000	22.9000

Summary Statistics on Data

TABLE 2 of 2

GRP	IDENTIFICATION	VARIANCE	SD	SEM	C.V. %
1	Control	8.6778	2.9458	0.9315	13.5751
2	5%	8.0000	2.8284	0.8944	9.4281
3	15%	115.3778	10.7414	3.3967	39.2022
4	30%	34.5444	5.8775	1.8586	21.8493
5	60%	24.7667	4.9766	1.5737	19.8271
6	100%	29.6556	5.4457	1.7221	23.7803

Steel's Many-One Rank Test			- Ho: Control<Treatment			
GROUP	IDENTIFICATION	MEAN IN ORIGINAL UNITS	RANK SUM	CRIT. VALUE	DF	SIG 0.01
1	Control	21.7000				
2	5%	30.0000	153.00	67.00	10.00	
3	15%	27.4000	140.50	67.00	10.00	
4	30%	26.9000	132.50	67.00	10.00	
5	60%	25.1000	132.00	67.00	10.00	
6	100%	22.9000	123.00	67.00	10.00	

Critical values are 1 tailed (k = 5)



PARAGON
Laboratories

Marh 28, 2011

Ms. Amy Vail
City of Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601

Subject: Executive Summary for Project No. 177739

Dear Ms. Vail:

Please find attached the biomonitoring report for testing conducted March 3, 2011 through March 10, 2011. The results for this evaluation are:

Test Species	TU _a	TU _c	MATC
<i>Ceriodaphnia dubia</i>	0	0	>100%

This exposure test was valid since it met the criteria for survival and reproduction in the controls. Should you have any questions pertaining to this report, please let me know.

Sincerely,

Jeannine Canze
Paragon Laboratories, Inc.-Aquatic Toxicology



MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY – WATER BUREAU
CERIODAPHNIA DUBIA CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

INSTRUCTIONS: Use this form to report chronic toxicity test results. Use separate forms for more than 1 test. Attach all raw data sheets to this report unless reporting for NPDES permit application.

1. NAME OF FACILITY (on NPDES permit) CADILLAC WWTP		2. NPDES PERMIT # M I 0 2 0 2 5 7					
3. RECEIVING WATER (as designated in permit) CLAM RIVER		4. OUTFALL 001					
5. RECEIVING WATER CONCENTRATION (if known)							
6. TEST LAB (Name and Address) PARAGON LABORATORIES, INC., 12649 RICHFIELD COURT, LIVONIA, MI 48150							
7. TEST START DATE 3/3/11	8. TEST END DATE 3/10/11	9. AGE RANGE OF ORGANISMS AT TEST START LESS THAN 24 HOURS	10. REPORT DATE 3/28/11				
11. NAME OF PERSON CONDUCTING TEST JEANNINE CANZE,		12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT PARAGON LABORATORIES, INC. - (734) 462 - 3900					
13. SAMPLE COLLECTION DATES Sample 1: 3/1/11 – 3/2/11 Sample 2: 3/3/11 3/4/11 Sample 3: 3/6/11 – 3/7/11		14. DATE RECEIVED Sample 1: 3/3/11 Sample 2: 3/5/11 Sample 3: 3/8/11					
15. ARRIVAL TEMP (°C) Sample 1: 0.5 Sample 2: 0.1 Sample 3: 2.3							
16. DATE OF FIRST USE Sample 1: 3/3/11 Sample 2: 3/5/11 Sample 3: 3/8/11		17. TOTAL RESIDUAL CHLORINE (mg/l) Sample 1: <0.02 Sample 2: <0.02 Sample 3: <0.02					
18. AMMONIA (mg/l as N) Sample 1: <0.5 Sample 2: <0.5 Sample 3: <0.5							
19. WAS SAMPLE DECHLORINATED? Sample 1: ☐ YES ☒ NO Sample 2: ☐ YES ☒ NO Sample 3: ☐ YES ☒ NO		20. DESCRIBE DECHLORINATION (if any)					
21. EFFLUENT SAMPLES WERE COLLECTED (check one) ☐ BEFORE CHLORINATION ☐ AFTER CHLORINATION ☐ AFTER CHLORINATION, BEFORE DECHLORINATION ☐ AFTER DECHLORINATION ☐ FACILITY DOES NOT CHLORINATE							
22. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded holding time.)							
23. EFFLUENT FILTERED? ☐ YES ☒ NO	24. STATE MESH SIZE OF FILTER (if filtered)						
25. EFFLUENT SAMPLE TYPE (check one type for each sample) Sample 1: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE Sample 2: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE Sample 3: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE			26. IDENTIFY THE DILUENT (O ₁) CONTROL MHRW IDENTIFY THE SECONDARY (O ₂) CONTROL (if used) N/A				
27. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCTION							
CONCENTRATION OF EFFLUENT (%)	O ₁	O ₂	5%	15%	30%	60%	100%
48-HOUR SURVIVAL (%)	100	N/A	100	100	100	100	100
7-DAY MEAN REPRODUCTION/FEMALE	18.7	N/A	20.1	20.1	19.5	24.4	23.1
7-DAY MEAN SURVIVAL (%)	100	N/A	100	100	100	100	100
28. 48-HOUR LC ₅₀ (%) >100%		29. TU _a (acute toxic units) 0					
30. 7-DAY CHRONIC VALUE (%) >100%	31. NOEC >100%		32. LOEC >100%		33. TU _c (chronic toxic units) 0		

EQP5945 (Rev. 6/2007)

TOXICITY EVALUATION
Sample Receipt Information

Client: Cadillac WWTP
NPDES Permit No.: MI0020257
Sample No.: 177739
Analyst: J. Canze

Sample	1	2	3
Sample Type	24 Hour Composite	24 Hour Composite	24 Hour Composite
Sample Date	3/1/11 - 3/2/11	3/3/11-3/4/11	3/6/11-3/7/11
Arrival Date/Time	3/3/2011 11:48	3/5/2011 10:57	3/8/2011 12:18

Parameter	Units	Arrival Chemistry		
Arrival Temperature	°C	0.5	0.1	2.3
Arrival pH	S.U.	7.4	7.7	7.3
Ammonia	mg/L	<0.5	<0.5	<0.5
Residual Chlorine	mg/L	<0.02	<0.02	<0.02
Alkalinity	mg/L CaCO ₃	110	110	90
Hardness	mg/L CaCO ₃	270	280	260

TOXICITY EVALUATION
Ceriodaphnia Survival and Reproduction Test

Client: Cadillac WWTP
Sample No. 177739
Analyst: J. Canze

Test Start Time: 3/3/11 3:00 PM
Test Stop Time: 3/10/11 3:30 PM

Effluent Concentration	Date	Day	Replicate I.D.										No. of Young	No. of Adults	Young per Adult
			A	B	C	D	E	F	G	H	I	J			
Moderately Hard water	3-Mar	0	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	4-Mar	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	5-Mar	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	6-Mar	3	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	7-Mar	4	2	2	2	3	4	2	3	2	3	3	26	10	2.6
	8-Mar	5	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	9-Mar	6	7	8	8	7	7	6	7	4	4	7	65	10	6.5
	10-Mar	7	12	10	13	8	10	8	8	10	9	8	96	10	9.6
	Total		21	20	23	18	21	16	18	16	16	18	187	10	18.7

Effluent Concentration	Date	Day	Replicate I.D.										No. of Young	No. of Adults	Young per Adult
			A	B	C	D	E	F	G	H	I	J			
5%	3-Mar	0	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	4-Mar	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	5-Mar	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	6-Mar	3	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	7-Mar	4	2	2	4	3	3	2	3	2	2	3	26	10	2.6
	8-Mar	5	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	9-Mar	6	7	0	10	8	7	10	6	4	4	9	65	10	6.5
	10-Mar	7	16	0	12	11	9	12	14	12	8	16	110	10	11.0
	Total		25	2	26	22	19	24	23	18	14	28	201	10	20.1

Effluent Concentration	Date	Day	Replicate I.D.										No. of Young	No. of Adults	Young per Adult
			A	B	C	D	E	F	G	H	I	J			
15%	3-Mar	0	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	4-Mar	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	5-Mar	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	6-Mar	3	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	7-Mar	4	3	2	5	4	2	3	2	2	3	2	28	10	2.8
	8-Mar	5	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	9-Mar	6	6	7	8	7	6	9	7	6	7	3	66	10	6.6
	10-Mar	7	13	8	15	8	10	8	13	13	14	5	107	10	10.7
	Total		22	17	28	19	18	20	22	21	24	10	201	10	20.1

TOXICITY EVALUATION
Ceriodaphnia Survival and Reproduction Test

Client: Cadillac WWTP
Sample No. 177739
Analyst: J. Canze

Test Start Time: 3/3/11 3:00 PM
Test Stop Time: 3/10/11 3:30 PM

Effluent Concentration	Date	Day	Replicate I.D.										No. of Young	No. of Adults	Young per Adult
			A	B	C	D	E	F	G	H	I	J			
30%	3-Mar	0	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	4-Mar	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	5-Mar	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	6-Mar	3	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	7-Mar	4													
			3	5	3	2	4	3	4	4	0	3	31	10	3.1
	8-Mar	5													
			0	0	0	0	0	0	0	0	0	0	0	10	0.0
	9-Mar	6													
			7	8	6	4	9	11	6	6	0	6	63	10	6.3
	10-Mar	7													
			17	9	15	3	12	16	13	8	0	8	101	10	10.1
	Total		27	22	24	9	25	30	23	18	0	17	195	10	19.5

Effluent Concentration	Date	Day	Replicate I.D.										No. of Young	No. of Adults	Young per Adult
			A	B	C	D	E	F	G	H	I	J			
60%	3-Mar	0	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	4-Mar	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	5-Mar	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	6-Mar	3	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	7-Mar	4													
			5	4	3	4	2	6	4	4	5	2	39	10	3.9
	8-Mar	5													
			0	0	0	0	0	0	0	0	0	0	0	10	0.0
	9-Mar	6													
			8	10	10	7	6	9	7	8	8	8	81	10	8.1
	10-Mar	7													
			15	10	11	13	11	14	12	12	15	11	124	10	12.4
	Total		28	24	24	24	19	29	23	24	28	21	244	10	24.4

Effluent Concentration	Date	Day	Replicate I.D.										No. of Young	No. of Adults	Young per Adult
			A	B	C	D	E	F	G	H	I	J			
100%	3-Mar	0	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	4-Mar	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	5-Mar	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	6-Mar	3	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	7-Mar	4													
			3	4	2	4	5	4	2	4	3	2	33	10	3.3
	8-Mar	5													
			0	0	0	0	0	0	0	0	0	0	0	10	0.0
	9-Mar	6													
			7	10	8	10	8	8	6	9	6	7	79	10	7.9
	10-Mar	7													
			15	17	10	16	9	11	8	15	3	15	119	10	11.9
	Total		25	31	20	30	22	23	16	28	12	24	231	10	23.1

TOXICITY EVALUATION
Ceriodaphnia Survival and Reproduction Test
 Routine Chemical and Physical Determination

Client: Cadillac WWTP

Sample No. 177739

Analyst: J. Canze

Test Start Time: 3/3/11 3:00 PM

Test Stop Time: 3/10/11 3:30 PM

Parameter	Units	Day						
		3-Mar	4-Mar	5-Mar	6-Mar	7-Mar	8-Mar	9-Mar
		0	1	2	3	4	5	6

Effluent Concentration:	Moderately Hard Water							
Temperature	°C	25	25	25	25	25	25	25
DO, Initial	mg/L	8.1	8.1	8.5	7.5	8.7	7.9	8.3
DO, Final	mg/L	8.3	8.7	8.4	8.3	8.2	8.2	8.1
pH, Initial	S.U.	8.2	8.0	8.1	8.1	8.2	8.1	8.0
pH, Final	S.U.	8.2	8.1	8.1	8.0	8.1	8.1	8.1
Conductivity, Initial	umhos/cm	310	310	300	310	300	310	300
Conductivity, Final	umhos/cm	350	320	350	330	340	340	340

Effluent Concentration:	5%							
Temperature	°C	25	25	25	25	25	25	25
DO, Initial	mg/L	8.0	8.4	7.9	7.5	8.6	7.9	8.1
DO, Final	mg/L	8.2	8.7	8.4	8.3	8.5	8.0	8.1
pH, Initial	S.U.	8.1	8.0	8.1	8.0	8.1	8.1	8.0
pH, Final	S.U.	8.2	8.1	8.2	8.0	8.1	8.1	8.2
Conductivity, Initial	umhos/cm	350	350	360	380	350	350	350
Conductivity, Final	umhos/cm	410	360	410	420	410	390	400

Effluent Concentration:	15%							
Temperature	°C	25	25	25	25	25	25	25
DO, Initial	mg/L	8.0	8.4	8.5	7.8	8.6	8.2	8.1
DO, Final	mg/L	8.2	8.7	8.4	8.3	8.4	8.0	8.3
pH, Initial	S.U.	8.0	8.0	7.9	7.9	8.0	8.1	8.0
pH, Final	S.U.	8.2	8.1	8.2	8.0	8.1	8.1	8.0
Conductivity, Initial	umhos/cm	440	450	500	520	430	450	420
Conductivity, Final	umhos/cm	520	460	580	570	500	480	500

Effluent Concentration:	30%							
Temperature	°C	25	25	25	25	25	25	25
DO, Initial	mg/L	8.1	8.5	8.5	8.2	8.6	8.2	8.1
DO, Final	mg/L	8.2	8.7	8.4	8.4	8.5	8.3	8.5
pH, Initial	S.U.	8.0	7.9	7.9	7.8	7.9	8.0	7.9
pH, Final	S.U.	8.2	8.1	8.3	8.0	8.2	8.1	8.2
Conductivity, Initial	umhos/cm	590	620	590	660	620	590	560
Conductivity, Final	umhos/cm	670	630	700	740	730	640	720

Effluent Concentration:	60%							
Temperature	°C	25	25	25	25	25	25	25
DO, Initial	mg/L	8.2	8.7	9.0	8.3	8.5	8.2	8.1
DO, Final	mg/L	8.3	8.6	8.4	8.4	8.5	8.2	8.6
pH, Initial	S.U.	7.9	7.8	7.8	7.8	7.8	8.0	7.9
pH, Final	S.U.	8.3	8.1	8.3	8.1	8.2	8.1	8.2
Conductivity, Initial	umhos/cm	860	930	790	820	900	850	830
Conductivity, Final	umhos/cm	970	920	960	930	1050	910	990

Effluent Concentration:	100%							
Temperature	°C	25	25	25	25	25	25	25
DO, Initial	mg/L	8.1	8.8	9.5	8.4	8.5	8.2	8.1
DO, Final	mg/L	8.3	8.6	8.4	8.3	8.4	8.2	8.6
pH, Initial	S.U.	7.8	7.8	7.7	7.6	7.8	7.9	7.9
pH, Final	S.U.	8.3	8.2	8.3	8.1	8.3	8.2	8.3
Conductivity, Initial	umhos/cm	1250	1320	1280	1300	1290	1240	1220
Conductivity, Final	umhos/cm	1380	1400	1560	1420	1530	1330	1490

RC177739

Title: Ceriodaphnia Reproduction Cadillac WWTP 177739
 File: cr177739 Transform: NO TRANSFORMATION

Kolmogorov Test for Normality

D = 0.1436 (p-value = 0.0036)
 D* = 1.1266

Critical D* = 1.035 (alpha = 0.01 , N = 60)
 = 0.895 (alpha = 0.05 , N = 60)

Data FAIL normality test (alpha = 0.01). Try another transformation.

Warning - The first three homogeneity tests are sensitive to non-normality and should not be performed with this data as is.

Summary Statistics on Data

TABLE 1 of 2

GRP	IDENTIFICATION	N	MIN	MAX	MEAN
1	Control	10	16.0000	23.0000	18.7000
2	5%	10	2.0000	28.0000	20.1000
3	15%	10	10.0000	28.0000	20.1000
4	30%	10	0.0000	30.0000	19.5000
5	60%	10	19.0000	29.0000	24.4000
6	100%	10	12.0000	31.0000	23.1000

Summary Statistics on Data

TABLE 2 of 2

GRP	IDENTIFICATION	VARIANCE	SD	SEM	C.V. %
1	Control	6.0111	2.4518	0.7753	13.1110
2	5%	57.6556	7.5931	2.4012	37.7767
3	15%	22.5444	4.7481	1.5015	23.6224
4	30%	81.6111	9.0339	2.8568	46.3276
5	60%	10.0444	3.1693	1.0022	12.9889
6	100%	35.8778	5.9898	1.8941	25.9299

Steel's Many-One Rank Test

- Ho: Control < Treatment

GROUP	IDENTIFICATION	MEAN IN ORIGINAL UNITS	RANK SUM	CRIT. VALUE	DF	SIG 0.01
1	Control	18.7000				
2	5%	20.1000	124.00	67.00	10.00	
3	15%	20.1000	121.00	67.00	10.00	
4	30%	19.5000	121.00	67.00	10.00	
5	60%	24.4000	148.50	67.00	10.00	

Page 1

			KC177739			
6	100%	23.1000	131.50	67.00	10.00	

Critical values are 1 tailed (k = 5)